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Assessing the synergies between textile sludge and low-density polyethylene (LDPE) Co-pyrolysis: Experimental insights and machine learning

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ABSTRACT

Co-pyrolysis leverages complementary properties of diverse feedstock to improve conversion efficiency, offering a sustainable route for integrated waste management and energy production. This study investigates reaction kinetics, synergistic interactions, product analysis, and machine learning prediction for co-pyrolyzing textile sludge (TS) with low-density polyethylene (LDPE) at mass ratios of 25%TS:75%LDPE, 50%TS:50%LDPE, and 75%TS:25%LDPE. Thermogravimetric analysis was performed from room temperature to 1000 °C at 2.5, 5, 7.5, and 10 °C/min. Model-free methods like linear differential Friedman, linear integral Kissinger-Akahira-Sunose (KAS), and Ozawa-Flynn-Wall (OFW) were used to evaluate kinetic parameters. Average E_a (Friedman) was 262.61 kJ/mol for 25%TS:75%LDPE, 178.34 kJ/mol for 50%TS:50%LDPE and 267.80 kJ/mol for 75%TS:25% LDPE. Furthermore, positive synergistic interactions were most significant between 450–600 °C with the dominant peaks at 500 °C for all blended samples. Fixed-bed co-pyrolysis of the optimum blend (50%TS:50% LDPE) at 500 °C produced 22% pyro-oil, 34% biochar, and 44% gaseous products. Moreover, obtained pyro-oil comprises of hydrocarbons and long-chain aliphatic derivatives, N-containing heterocycles and amines, carboxylic and phenolic acids, ethers and acetal, phthalates, and the carbohydrate derivatives. Furthermore, machine learning models like Artificial neural networks (ANNs), Classification & regression trees (C&RT), and support vector machine (SVM) were developed to predict E_a. ANN performed best for 50%TS:50%LDPE and 75% TS:25%LDPE (R^2 = 0.999 and 0.997) while C&RT excelled for 25%TS:75%LDPE (R^2 = 0.988). Findings demonstrate pronounced synergistic interactions and integrated kinetic and machine learning prediction strategy for converting diverse waste into energy.

1. Introduction

Rapid industrial growth has exacerbated the challenges associated with disposing of sewage, wastewater, and sludge (Bibby and Peccia, 2013). Textile sludge (TS), a byproduct of the textile industry, poses significant environmental challenges due to its content of pathogenic bacteria, organic pollutants and toxic substances such as heavy metals and dyes. It is estimated that for every million tons of textile wastewater treated, approximately 25 m^3 of the sludge is generated, with two-thirds of this sludge having modifiable characteristics (Huang et al., 2022). Due to its high energy and nutrient content, TS offers a significant opportunity for energy and resource recovery within circular economies

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(Huang et al., 2018). Effectively managing TS degradation is crucial for achieving environmental sustainability. Presently, several technologies specifically designed for municipal sludge treatment are being used for TS treatment. Nevertheless, traditional disposal methods such as landfilling, composting, and agricultural application have notable limitations and disadvantages (Xie et al., 2018; Khalid et al.,). Consequently, these methods often prove inadequate for effective sludge management, resulting in significant environmental issues. Therefore, it is crucial to investigate other methods of disposal and safe handling of sludge waste. One promising way that has garnered interest is the conversion of sludge into the fuel is by thermochemical processes like the pyrolysis, which offers a promising solution for managing the TS by reducing waste volume and pollution risks, while being more environmentally friendly compared to other thermochemical methods (Park et al., 2010; Karthikeyan et al., 2026). Thermogravimetric analysis (TGA) is widely utilized for investigating the kinetics of thermal degradation in solid materials (Mehdi et al., 2026). Pyrolysis process generates valuable products, including non-condensable gases, pyro-oil, and biochar, which can be further processed into fuels or used as chemical products, adsorbents, or catalysts (Wang et al., 2020a; State et al., 2021). However, the high ash content and the low calorific value of TS hinder its effectiveness when undergoing pyrolysis alone (Urych and Smoliński, 2021). Therefore, by combining the TS with second-generation feedstocks, such as biomass or polymers, the pyrolysis process can be enhanced and product quality improved (Zhang et al., 2020a).

Waste plastics are particularly promising for co-pyrolysis due to their high potential for resource reuse. Polyolefin plastics, including Low-Density Polyethylene (LDPE), High-Density Polyethylene (HDPE), and Polypropylene (PP), account for approximately 50% of the global plastic consumption as a type of polymer material (Zheng and Suh, 2019). It has been investigated that globally ~19–23 million tons of plastic waste is generated every year. LDPE has the significant portion of this waste because of its extensive application in the packaging material along with its limited recyclability by conventional methods (Tiago et al., 2023; Zaker et al., 2021a). Moreover, LDPE poses serious risks to both aquatic and terrestrial ecosystems, as it can transport pathogens, organic contaminants, heavy metals, and hazardous chemicals (Ali et al., 2021). Therefore, LDPE is an excellent feedstock for the cop-pyrolysis with TS because of its higher calorific value and a greater effective hydrogen-to-carbon ratio (H/C), and lower levels of ash and oxygen compared to biomass. This approach not only reduces waste but also provides thermal disinfection by leveraging the compatibility of waste feedstocks to create higher-value products while minimizing environmental impact (Raheem et al., 2018).

To enhance predictive analysis for co-pyrolysis kinetic studies, predictive modeling has been employed like the Artificial Neural Networks (ANNs), Classification & Regression Trees (C&RT), and Support Vector Machines (SVM). ANNs are utilized for predicting kinetic parameters but may struggle with overfitting when dealing with complex datasets (Hassija et al., 2024). The C&RT model excels due to its interpretability, providing clear insights into decision-making processes while effectively managing nonlinear relationships and the missing data (Breiman, 2017). SVM are supervised learning algorithms that classify data by finding an optimal hyperplane to maximize the distance between classes. It is applied to the continuous outcome problems, offering a robust tool for modeling nonlinear kinetic behaviors.

Previous studies have examined in isolation the pyrolysis kinetics of TS (Wahab et al., 2022) and LDPE (Kple et al., 2017), as well as the co-pyrolysis of TS and medical plastics (Ding et al., 2021). Additionally, the co-pyrolysis of LDPE and biomass (Yunnan Pine) (Zheng et al., 2018) and the co-pyrolysis of sewage sludge and LDPE (Zaker et al., 2021b; Ma et al., 2024) have also been explored in previous studies. Furthermore, catalytic co-pyrolysis of biomass (empty fruit bunch) with polypropylene over NiFe/ZSM−5catalyst were also investigated where optimum blend achieve the higher liquid yield (Valentino et al., 2026). However, there has been no research reported on co-pyrolyzing TS with LDPE. Moreover, there is a significant gap in previous studies in applying machine learning approaches which include ANN and C&RT along with SVM for activation energy (E_a) prediction in co-pyrolytic degradation. Therefore, this study marks the first attempt to investigate co-pyrolysis kinetics of TS with LDPE using model-free methods along with the integration of advance machine learning models in predicting the E_a. Furthermore, synergistic interactions between TS and LDPE are identified through statistical analysis such as Principal Component Analysis (PCA). Additionally, an optimum blend based on the average E_a was selected for co-pyrolysis in a fixed-bed reactor to analyze the products generated from the co-pyrolytic breakdown of TS and LDPE. The integration of model-free kinetics with machine learning approaches is crucial for modeling complex relationships as it somehow enables the effective analysis of intricate reaction processes, provides valuable insights and the enhances predictive accuracy. Thus, this work aims to integrate the kinetic parameters and machine learning prediction which enhances resource efficiency and sustainability, leveraging data-driven approaches to optimize processes and developing sustainable solutions.

2. Materials and methods

The TS used for this work was sourced from a textile manufacturing facility which is present in Multan, Pakistan. At the start, the TS was dried under sun for 5 days to remove the moisture present on the surface, and then it was dried in an oven (101–1AB) at 110 ºC for approximately 24 h to remove the inbound moisture present in the TS.

In contrast, LDPE employed in this work consists of beads which were obtained from a local supplier. Beads were initially washed to remove any impurity and then dried. After drying, both TS and beads of LDPE converted into fine powders by a laboratory scale grinder. The obtained powders were then passed through (DIN−4188) 0.2 mm screen. After that powdered sieved samples of TS and LDPE were employed for blend preparation.

2.1. Textile sludge and low-density polyethylene blend preparation

The blends for co-pyrolysis were made by mixing different ratios of TS with LDPE denoted as 25%TS:75%LDPE, 50%TS:50%LDPE and 75% TS:25%LDPE, respectively. These specific blends were chosen to systematically investigate the effect of different compositions of TS and LDPE on co-pyrolysis behavior. A digital GSM balance was utilized to weigh the samples for precision and accuracy, followed by mixing with a vortex mixer for 5–10 min to ensure uniformity in the blends. Fig. 1 shows the schematic of TS and LDPE preparation from the initial stage to the powders form and its blends for co-pyrolytic process.

The ultimate analysis which includes Carbon (C), Hydrogen (H), Nitrogen (N) and Sulfur (S) were determined from elemental analyzer (ECS 8020) using nitrogen gas. Furthermore, proximate analysis such as, moisture, volatile matter, fixed carbon and ash were determined by following the standard method of American Society of Testing Materials (ASTM). The moisture content was calculated by drying the samples using drying oven while the volatile matter and ash were calculated using the equipment muffle furnace (SX−2.5–12). The gross calorific value (GCV) of the TS, LDPE, 25%TS:75%LDPE, 50%TS:50%LDPE, and 75%TS:25%LDPE were determined using bomb calorimeter (PAR−6200) where the sample was combusted in the calorimetric bomb. Approximately 0.5–0.6 g of each sample was put in a crucible and sealed within the oxygen-filled chamber. The samples were ignited, and GCV was determined by comparing the heat release with that of a calibration standard. Moreover, the functional groups present in TS, LDPE, 25%TS:75%LDPE, 50%TS:50%LDPE, and 75%TS:25%LDPE were analyzed using a FTIR spectrophotometer (Perkin Elmer Spectrum 100) across a range of 400–4000 cm^{-1}. For FTIR analysis, solid sample pellets were made by using potassium bromide (KBr) and analyzed by using an FTIR spectrophotometer that works by directing infrared light onto the

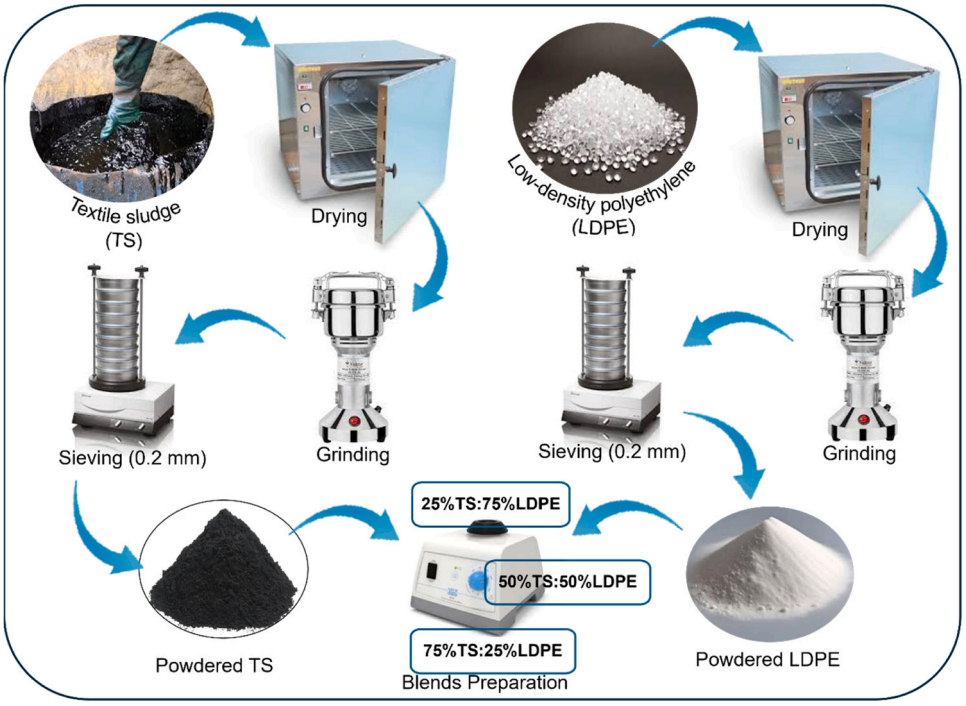


Fig. 1. Schematics of blend preparation of TS with LDPE.

sample and measuring the specific frequencies, which helps to identify the sample molecular vibrations and functional groups.

2.2. Thermal analysis

The samples of TS, LDPE, 25% TS:75% LDPE, 50% TS:50% LDPE, and 75% TS:25% LDPE were examined by using the TGA (Mettler Toledo). Approximately 8–10 mg of the powdered samples were placed in a crucible and is heated from 25 to 1000 °C under a continuous nitrogen flow of 50 mL/min. The analysis were performed at four different heating rates of 2.5, 5, 7.5, and 10 °C/min. All the experiments were repeated two times to ensure repeatability of the results. The data which was collected on mass loss, temperature and the time were then used to determine kinetic parameters. The selection of heating rates and experimental procedures followed the recommendations of the ICTAC Kinetics Committee, ensuring reliable kinetic data for thermal decomposition analysis (Vyazovkin et al., 2020).

2.2.1. Kinetic analysis

The kinetic study was performed by using various model-free methods that eliminate the need for predefined reaction mechanism. Equations for the kinetics and model-free methods are provided in supplementary data from [Eq. (1**s**) to Eq. (6**s**)]. The E_a is calculated for the linear differential (Friedman) by the graph plotting between ln(dα/dt) against 1/T as shown in [Eq. 4**s**]. In addition to that, Kissinger-Akahira-Sunose (KAS), a linear integral approach offers another way to evaluate kinetic parameters by plotting the graph between $\ln(\beta/T^2)$ and the 1/T which is shown by [Eq. 5**s**] present in supplementary materials. The other integral method known as Ozawa-Flynn-Wall (OFW) derives E_a from the slope of a plot of ln β versus the reciprocal of temperature (1/T) which is presented in supplementary data as [Eq. 6**s**]. Utilizing these methods provides valuable insights into the thermochemical characteristics for the pyrolysis and co-pyrolysis of all samples.

2.2.2. Synergy index

The interaction between the TS and LDPE during co-pyrolysis was assessed by examining the deviation conversion difference (ΔC) between the experimental conversion (C_{Exp}), and the calculated conversion value (C_{Cal}), which is shown in [Eqs. (1–2)]

$$\Delta C = C_{Exp} - C_{Cal} \quad (1)$$

$$C_{Cal} = \omega_{TS}.C_{TS,\ Exp} + \omega_{LDPE}.C_{LDPE,Exp} \quad (2)$$

In the [Eq. 2], $C_{TS,Exp}$ and the $C_{LDPE,Exp}$ represent the experimental conversion values for TS and LDPE being pyrolyzed individually. The symbols ω_{TS} and the ω_{LDPE} indicate the mass fractions of TS and LDPE in the blend, respectively.

If ΔC value is close to zero, it suggests that there is no interaction between the components during co-pyrolysis. A negative ΔC means that the combined material has high theoretical value (calculated) than the experimental result, indicating an inhibitory effect. Conversely, a positive ΔC points to an increase in synergistic effect (Peng et al., 2015a).

2.3. Statistical analysis

2.3.1. Principle component analysis

To systematically minimize the number of parameters involved in kinetic analysis while retaining most of the original variability, a principal component analysis (PCA) was conducted. In PCA, firstly, the variables are being normalized, followed by transforming these probable variables into the principal components i.e. linear combinations of the original parameters by identifying directions in data that exhibit high variance. The scores from these principal components (PC) facilitate visualization of data in the lower-dimensional space. PC scores represent the projections of experimental data points onto the PC, whereas PC loadings are the coefficients used to calculate these PC scores.

2.3.2. Hierarchal cluster analysis

Hierarchical cluster analysis (HCA) is a widely used statistical method for grouping objects exhibiting similar characteristics or features into clusters. The method is an unsupervised learning algorithm; therefore, it requires no pre-defined categories. It establishes inherent

groupings within the data using similarity or distance metrics. The HCA methodology is composed of various steps. Data preparation begins with the organization of data into a matrix structure, with subsequent normalization or standardization of variables. Following this, a distance matrix is generated to determine the pair-wise distances between each object within the dataset. Then, the hierarchical clustering process employs either an agglomerative (bottom-up) or divisive (top-down) methodology. In the common agglomerative clustering approach, each object initially represents a distinct cluster. The algorithm proceeds by iteratively combining the closest clusters until a single, unified cluster is obtained. Subsequently, an optimal number of clusters is determined through the analysis of a dendrogram, a hierarchical tree diagram that illustrates the cluster relationships.

2.4. Co-pyrolysis experiment

The schematic for co-pyrolysis is depicted in [Fig. 1**s**], which is presented in the supplementary data. The experimental setup for co-pyrolysis is basically the fixed bed reactor. The detailed reactor configuration is present in our previous work (Khan et al., 2026). The reactor is made from the stainless-steel tube which has outer diameter of 14 mm along with 12 mm inner diameter. Powdered sample was put in the tube middle part which is insulated by the quartz wool. Heating was applied via an electric heater to which temperature controller is connected, that allows precise temperature regulation. To control the gas flow, mass flow controller (model MF (4603-N1, 1), by Servoflo Corporation, USA), integrated into the system. The nitrogen gas flow was set at 50 mL/min. The heating rate during co-pyrolysis was maintained at 10 °C per minute. Gaseous products (non-condensable gases) generated were collected after condensation from the condenser and the pyro-oil obtained in the other vessel. The co-pyrolysis experiment was repeated twice and the average values for the yield (pyro-oil, biochar, non-condensable gases) were taken.

2.4.1. Co-pyrolysis products characterization

The non-condensable gases generated during the co-pyrolysis process were collected from downstream of the condenser which was examined by Gas Chromatograph with model (GC Pro, SHIMADZU, Japan). The analysis utilized a TCD (RT-MS5A column, 30 m × 0.32 mm), maintained at 200 °C, with a total time of 8 min. Nitrogen gas was used as a carrier gas, and calibration was performed by using standard gases with a purity of 99.9%. For each test, approximately 30 mL of the gas sample was collected at 500 °C at intervals of 20 min to evaluate the gas composition.

Pyro-oil GCV was obtained by using a bomb calorimeter to examine its potential as fuel. Additionally, FTIR spectroscopy was also employed to retrieve the various functional groups in pyro-oil that are obtained from the co-pyrolysis process. Further compositional analysis was conducted using a GC-MS system (Shimadzu QP2020 NX, Japan) which is equipped with the DB−5MS capillary column (30 m × 0.25 mm i.d., 0.25 μm film thickness). Helium served as the carrier gas in the system, and the pyro-oil sample mixed in n-hexane was injected with a 1:20 split ratio. The column temperature increased from 50 °C to 150 °C at 5 °C/min and after that increased from 150 °C to 290 °C with the rate of 10 °C/min. Both the detector and injector temperatures were set at the 290 °C. At the electron impact potential of 70 eV and the scanning range of 35–500 amu the spectra were recorded. Compound identification was achieved by matching the mass spectra against the National Institute of Standards and Technology (NIST) database.

The biochar obtained from co-pyrolysis was characterized by measuring its GCV to determine heating value, examining its morphology by the scanning electron microscopy (SEM), and analyzing its functional groups present using FTIR spectroscopy.

The pyro-oil and biochar yield obtained from the co-pyrolysis process is calculated by using the [Eq. 3] . However, yield of the gases (non-condensable) was calculated from mass balance by subtracting the yields of pyro-oil and biochar.

$$\text{Product yield}(\%) = \frac{W_p}{W_f} \times 100 \tag{3}$$

Pyro-oil and biochar weight (g) obtained from co-pyrolysis is denoted as W_p, while W_f represents the weight of the feedstock that is used in the co-pyrolysis process.

2.5. Machine learning topology

Statistica was employed for machine learning modeling. Various learning models includes the ANN, C&RT, and SVM were applied for E_a prediction. The ANN is composed of input layer, hidden layer along with the output layer. For the regression analysis, data flows from input layer to the hidden layer and then finally proceeds to the output layer. However, neurons in the hidden and the output layers are interconnected, with model errors reduced through the iterative adjustments of neuron weights and transfer functions. ANN makes use of the both the input and the target datasets to develop the linear relationship between the different variables (Mishra et al., 2023). For E_a prediction, input neurons consisted of the conversion (-), heating rate (°C/min) and temperature (K) where the output variable is the E_a. Additionally, dataset for training consisted of 70%, while the 15% is set for the testing purpose and remainimg 15% utilized for the validation. By Broyden-Fletcher-Goldfarb-Shanno (BFGS), regression analysis was conducted. Different activation functions were applied to both the hidden and the output layers which include the SoftMax, sine, logistic, hyperbolic, Gaussian, and the exponential functions. Each neuron is connected to the nodes which are being linked with output layer by the assigned weights and biases. The most suitable model is identified by use of the Automated Network Search (ANS) function. In addition, this work also employed the C&RT model, which generates the decision trees (DT) using iteratively dividing the dataset into subsets according to defined splitting criteria (Fávero et al., 2023). For classification tasks, the method uses criteria like Gini Index or the entropy to reduce data impurity, whereas for regression tasks it aims to minimize Mean Squared Error (MSE). DT is constructed by repeatedly choosing the most relevant features and optimal split points, producing the internal nodes that denote decision rules and leaf nodes that deliver the final output predictions. Additionally, the other model employed is the SVM which is supervised learning algorithm, finds an optimal hyperplane for separating the data into distinct classes. It maximizes the margin between the classes, focusing on the support vectors data points closest to the hyperplane. SVM can handle the non-linear problems by using kernel functions, making them effective for robust classification with minimal overfitting.

The assessment of different machine learning models performance was carried out by various metrics like the R^2, MSE, Mean Absolute Error (MAE), and the Root Mean Squared Error (RMSE). Performance metrices were assessed using the formulas to evaluate the ANN, C&RT and the SVM models. Different performance metrics equations are presented as [Eqs. (4–7)] (Ali et al., 2023).

$$R^2 = 1 - \frac{\sum_{i=1}^{N-1}\left[E_{a_i}^{\ m} - E_{a_i}^{\ p}\right]^2}{\sum_{i=1}^{N-1}\left[E_{a_i}^{\ m} - \overline{E_{a_i}^{\ m}}\right]^2} \tag{4}$$

$$MAE = \frac{1}{N}\left[\sum_{i=1}^{N-1}\left|E_{a_i}^{\ m} - E_{a_i}^{\ p}\right|\right] \tag{5}$$

$$MSE = \frac{1}{N}\left[\sum_{i=1}^{N-1}\left[E_{a_i}^{\ m} - E_{a_i}^{\ p}\right]^2\right] \tag{6}$$

$$RMSE = \sqrt{\frac{1}{N}\left[\sum_{i=1}^{N-1}\left[E_{a_i}^{\ m} - E_{a_i}^{\ p}\right]^2\right]} \tag{7}$$

In the above equations, N denotes the total observation numbers. Predicted value of E_a is denoted as E_{ai}^{p}, while the experimentally obtained (measured) value of the E_a is shown as E_{ai}^{m}, with $\overline{E_{ai}^{m}}$ indicates the mean of the experimentally obtained E_a. Employed models are considered good fit when the value of R^2 is close to 1 along with minimum error metrices. According to the literature, the value of R^2 above 0.85 generally indicates strong predictive performance (Tariq et al., 2022).

3. Results and the discussion

3.1. Physiochemical properties along with characterization

Different techniques such as ultimate analysis, proximate analysis, GCV and the FTIR spectroscopy are commonly used to evaluate the different properties of the samples. A comprehensive characterization like the ultimate analysis, proximate analysis and GCV of pure TS and LDPE were reported in our previous study (Khan et al., 2026, 2025). The ultimate analysis of TS reveals that it contains 16.4 wt% carbon, hydrogen content was 2.0 wt%, 2.2 wt% nitrogen while sulfur content measured as 3.1 wt%. More to that, proximate analysis of TS identifies the 3.3 wt% of the moisture, volatile matter accounting for 46.2 wt%, 2.5 wt% fixed carbon, and ash content observed as 48%. Additionally, for the LDPE, ultimate analysis shows 85.17 wt% carbon and 14.83 wt% hydrogen, while the proximate analysis shows that the LDPE contains the 100 wt% volatile matter. However, for 25%TS:75%LDPE, the ultimate analysis reveals that it has 67.98 wt% carbon, 11.62 wt% hydrogen while the nitrogen and sulfur was obtained as 0.56 wt% and 0.79 wt% respectively. Similarly, for 50%TS:50%LDPE, the ultimate analysis identifies 50.79 wt% carbon, 8.42 wt% hydrogen, 1.12 wt% nitrogen and 1.55 wt% sulfur. However, for the 75%TS:25%LDPE the ultimate analysis depicts that it has 33.6 wt% carbon, 5.2 wt% hydrogen while the nitrogen was found to be 1.65 wt% and sulfur content was obtained as 2.32 wt%. Moreover, the result of proximate analysis reveals that the moisture content was found to be 0.9 wt% for 25%TS:75% LDPE, while for 50%TS:50%LDPE it was obtained as 1.66 wt%. Volatile matter was obtained as 86.48 wt% and 73.09 wt%. Similarly, the findings retrieve the ash content as 12 wt% and 24 wt% respectively for 25% TS:75%LDPE and 50%TS:50%LDPE samples. Furthermore, for the sample 75% TS:25% LDPE, the moisture content was observed to be 2.33 wt%, while the volatile matter and ash were obtained at 58.66 wt% and 37.02 wt%, respectively. Additionally, the fixed carbon recoveries as 0.62 wt%, 1.26 wt% and 1.99 wt% for 25%TS:75%LDPE, 50% TS:50%LDPE and 75%TS:25%LDPE respectively.

Furthermore, the GCV of TS, LDPE, 25%TS:75%LDPE, 50%TS:50% LDPE and 75%TS:25%LDPE were recorded as 7.63 MJ/kg, 49.752 MJ/kg, 38 MJ/kg, 28.54 MJ/kg, 18.836 MJ/kg respectively. In the previous studies GCV of the TS was found as 8.64 MJ/kg (Gadhi et al., 2024) while for the LDPE the lower heating value was reported as 42.88 MJ/kg (Chaudhary et al., 2023). Table 1 depicts the results of ultimate analysis, proximate analysis and GCV of TS, LDPE, 25%TS:75%LDPE, 50% TS:50%LDPE and 75%TS:25%LDPE.

Fig. 2 shows the FTIR spectrum of all the samples. The spectra obtained from FTIR analysis reveals the presence of the various characteristic functional groups. The detailed FTIR analysis of pure TS and LDPE was already present in our previous research work (Khan et al., 2026, 2025). For the LDPE, the bands was observed between 2850–2920 cm^{-1} which is linked to -CH_2- stretching that forms the backbone of LDPE (Rajandas et al., 2012). However, the additional peaks at around 1466 and 723 cm^{-1} correspond to the bending and rocking of the methylene groups.

For the TS, peaks attributed to the stretching vibrations of hydroxyl groups (O–H) appear at 3480 cm^{-1}, while it is at 3423.40 cm^{-1} for 25% TS:75%LDPE, and 3429 cm^{-1} for 50%TS:50%LDPE. Additionally, for 75%TS:25%LDPE the peak was identified at 3431 cm^{-1}. The peaks shifts are majorly different as compared to pure TS. The (O-H) group presence shows the organic constituents presence which is very common because of various dyes used in TS (Bal, 1999). It has been observed that as the proportion of TS increases, slight shifts in the peak positions or the changes in peak intensities occurred. However, the progressive peak shifts with increasing the TS to the LDPE highlights those possible intermolecular interactions such as hydrogen bonding and physical mixing effects between TS and LDPE. The other prominent peaks observed between 1463.13 and 1655 cm^{-1} for the blended samples indicate the presence of -C═O stretching and -CH_2 bending deformation and the amides functional groups which are attributed to nitrogen containing organic compounds and the oxygenates that are found in TS (Wang et al., 2019). These findings confirm the heterogenous nature of TS and its interaction with LDPE.

3.2. Co-pyrolysis performance as a function of blending ratio and degradation stage

The thermal analysis of TS, LDPE, 25%TS:75%LDPE, 50%TS:50% LDPE and 75%TS:25%LDPE were conducted for its pyrolysis and co-pyrolysis degradation at four varying rates of 2.5 °C/min, 5 °C/min, 7.5 and 10 °C/min to examine the decomposition characteristics. Fig. 3 (A-B) shows the conversion and DTG profile of pure TS and LDPE pyrolysis. While the Fig. 4(A-C) illustrates a DTG plot of 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25%LDPE co-pyrolysis across these different heating rates.

DTG curves in Fig. 3**(A-B)** show the pyrolysis behavior of TS and LDPE. The detailed thermal analysis of pure TS and LDPE pyrolysis at varied heating rates is reported in our previous work (Khan et al., 2026, 2025). During TS pyrolysis, Stage I roughly around 25–215 °C for all stages which involves mass loss because of moisture and release of light volatiles. The overall mass loss at this stage is roughly in between 11.20% and 12.59% at all heating rates. Stage II (215–600 °C), for TS pyrolysis is the main decomposition phase as it causes the largest mass loss because of the degradation of proteins, fibers, and the other organics break down. Stage III (>600 °C) involves the slow breakdown of carbon-rich residues and the inorganic ash. For the Pyrolysis of LDPE, DTG curve shown in Fig. 3**(B)** shows that the major degradation of LDPE occurs in the active pyrolysis region which lies roughly in the temperature range of about 400 °C to 500 °C. During this temperature range, the polymer chains breakdown is most intense (Zaker et al., 2021b). Degradation of LDPE takes place in a single step. The mass loss of LDPE occurs because of breakage of the polymer chain. The overall mass for the LDPE is found to be 100%.

The DTG graph presented in Fig. 4**(A-C)** highlights the co-pyrolysis behavior for 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25%

Table 1
Ultimate & Proximate analysis and GCV results of TS, LDPE, 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25%LDPE.

Sample	Carbon (wt%)	Hydrogen (wt%)	Nitrogen (wt%)	Sulfur (wt%)	Moisture (wt%)	Ash (wt%)	Volatile Matter (wt%)	Fixed Carbon (wt%)	GCV (MJ/kg)
TS	16.4	2	2.2	3.1	3.3	48	46.2	2.5	7.63
LDPE	85.17	14.83	0	0	0	0	100	0	49.72
25%TS:75%LDPE	67.98	11.62	0.56	0.79	0.9	12	86.48	0.62	38
50%TS:50%LDPE	50.79	8.42	1.12	1.55	1.66	24	73.09	1.26	28.54
75%TS:25%LDPE	33.6	5.20	1.65	2.32	2.33	37.02	58.66	1.99	18.83

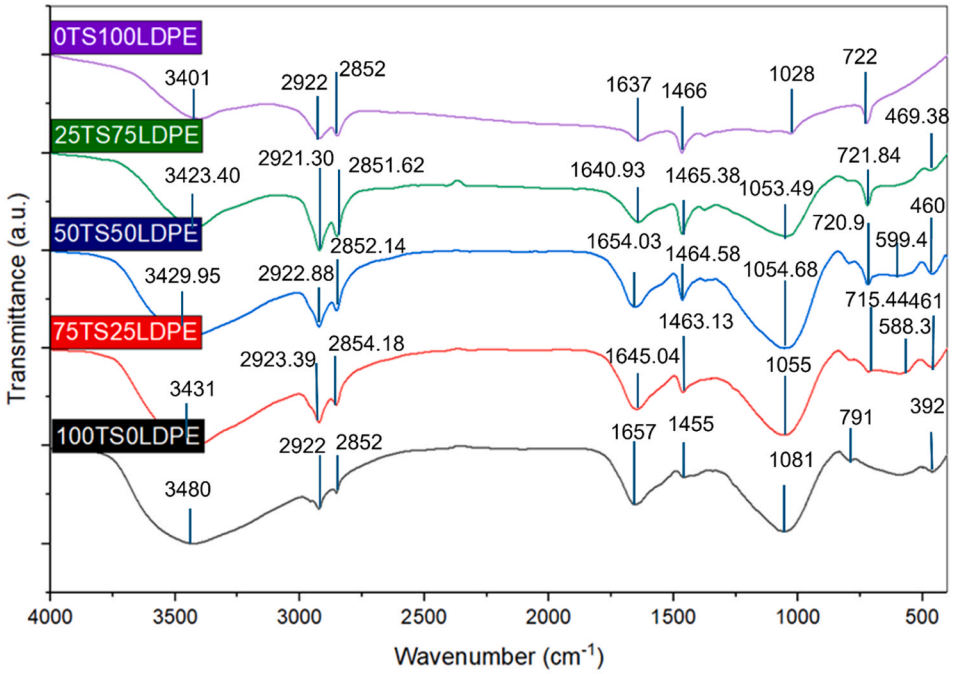


Fig. 2. FTIR spectrum of TS, LDPE, 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25%LDPE.

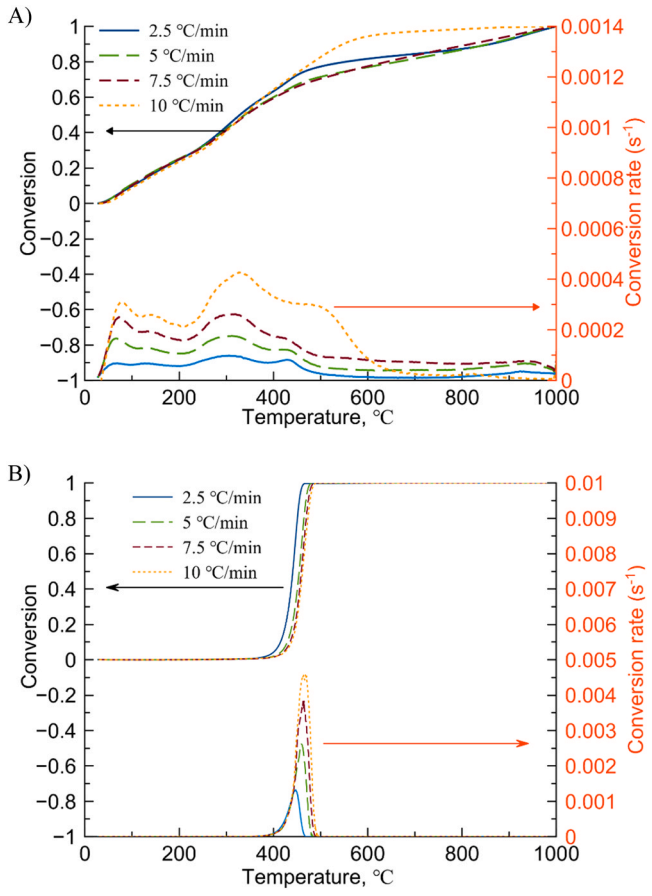


Fig. 3. Conversion and DTG curve of **A)** TS & **B)** LDPE at 2.5, 5, 7.5 and 10 °C/min.

LDPE, which unfolds in three distinct stages. Decomposition can be divided into moisture volatiles removal, organic matter breakdown due to TS, and inorganic transformation during these stages (Nguyen and Chiang, 2021). Stage-I retrieves the moisture removal from both surface water and the water trapped inside and the release of very light volatiles due to the TS fraction. Stage-II covers the release of further volatile substances through the gentle breakdown of TS which typically includes the breakdown of small organic compounds which include formic acid, saccharides, acetic acid, and alcohols, as well as larger organic materials like fibers, proteins, and saturated aliphatic chains found in the TS (Peng et al., 2015a, 2015b). These proteins are produced during the biological treatments that stabilize the material and the polymers formed throughout the sludge generation process. In addition to that, carbohydrates, cellulose, lipids, and other materials with low biodegradability could also break down during this decomposition stage. Additionally, Stage-III involves the breakdown of carbonaceous residues and inorganic materials, like the ash present in 25%TS:75% LDPE, 50%TS:50%LDPE and 75%TS:25%LDPE during its co-pyrolytic degradation. However, the secondary breakdown of different products, which include coke and tar, also occurs at this stage. Moreover, in this stage, non-extended summit results from the carbonization of a limited number of complex refractory organic substances, along with the gradual breakdown of stable inorganic materials (Ma et al., 2024). For the co-pyrolysis of 25%TS:75%LDPE as shown in Fig. 4**(A),** the Stage-I falls within the temperature from ambient to roughly at 400 °C. There is no very sharp peak observe in this range because of the increased fraction of LDPE in the blend. Few small spikes are observed in this region because of addition of small amount of TS. The active pyrolysis region roughly begins at approximately 400 °C and extends to around 500 °C. The maximum degradation peaks in this stage appears roughly around the temperature between 442 °C – 472 °C for all different heating rates. Overall mass loss recorded for 25%TS:75%LDPE are 85.02%, 87.68%, 86.9% and 86.56% for the four respective heating rates. Similarly for the co-pyrolysis of 50%TS:50%LDPE from Fig. 4 **(B)** the Stage-I starts from 25 °C and ends roughly at 200, 203.7, 217 and 220 °C at all four heating rates. However, Stage II starts roughly at 200–220 °C and ends at 500 °C. The active pyrolysis region curve will start to rise from 400 °C which dictates the interaction between TS and LDPE will occur and the degradation will start at this temperature. At the temperature roughly between 430 and 470 °C, four sharp peaks appear in this active

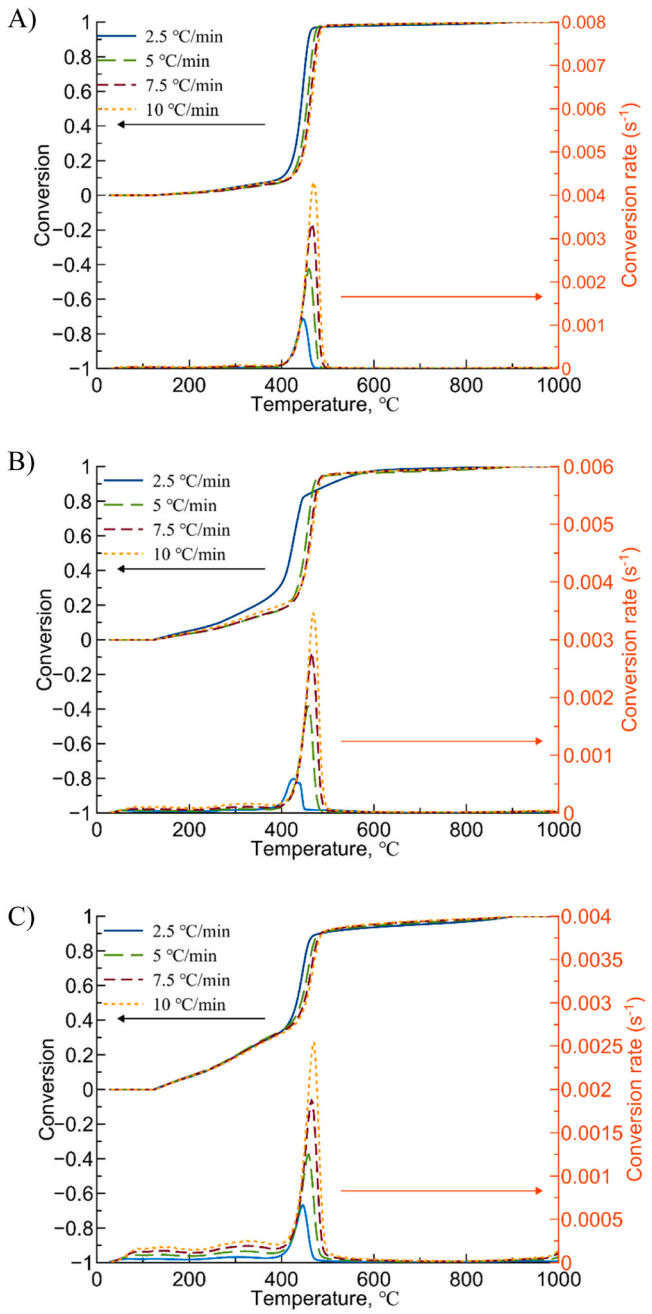


Fig. 4. Conversion and DTG curve of **A)** 25%TS:75%LDPE, **B)** 50%TS:50% LDPE & **C)** 75%TS:25%LDPE at four varied heating rates.

pyrolysis region for all the four heating rates. The major mass loss occurred at this stage; this is because of the breakdown of the macromolecule organic compound of the polyolefin materials (Singh et al., 2020; Yao et al., 2021). Stage III starts from 500 °C and ends when the asymptotic conversion reaches 1000 °C. The overall mass loss was found to be 73.64%, 71.14%, 74.33% and 72.79% respectively, at all four heating rates. Furthermore, for 75%TS:25%LDPE from Fig. 4**(C)**, Stage I roughly starts from ambient temperature to 200 °C where the removal of moisture and the release of light volatiles take place because of the TS present. Peaks at Stage-I are more profound in the 75%TS:25%LDPE as compared to 25%TS:75%LDPE and 50%TS:50%LDPE because of the addition of higher fraction of TS present. The overall mass loss obtained at this stage was 8.12%, 7.88%, 7.63%, 7.20% for all the four heating rates. Stage-II occurs from 200 °C and ends roughly at 500 °C where the maximum mass loss occurred. However, major degradation occurred roughly between 400 °C to 500 °C. The sharp peaks between this temperature range roughly appears between ~ 450–475 °C at all four different heating rates of 2.5, 5, 7.5 and 10 °C/min. The overall mass loss for 75%TS:25%LDPE was observed as 59.36%, 58.87%, 58.01% and 59.82% at all four heating rates. It can be seen that, among the blended samples, the overall mass loss for 75%TS:25%LDPE is lower. This is due to higher proportion of TS in the sample. Additionally, peaks are more dominant in 75%TS:25%LDPE due to increase TS content that is highly attributed to significant synergistic interactions between the TS and LDPE. Furthermore, It is observed that interaction between TS and LDPE advanced the co-pyrolysis process. Lignin and the cellulose along with the inorganic minerals present in the TS might have synergistically interacted with the LDPE (Ahmed et al., 2020). Additionally, synergistic interactions are also attributed to volatile compounds derived from TS, which contain oxygen-containing functional groups, and the alkane groups generated from the cracking of polymer material (Ding et al., 2021). In the prior work it has been demonstrated that co-pyrolysis of Polypropylene (P)P or Polyethylene (PE) with cellulose can produce aromatic compounds through the Diels-Alder reaction, while also suppressing the polymerization of oxygen-containing intermediates (Mullen et al., 2018). Table 2 shows the initial temperature, final temperature and weight loss during the co-pyrolysis of 25%TS:75%LDPE, 50% TS:50%LDPE and 75%TS:25%LDPE.

3.3. Kinetic parameters assessment

E_a is one of the parameter for the estimation of kinetic analysis, as it represents the minimum energy being required for chemical bonds breakage and initiate the reaction.

3.3.1. Activation energy

Model-free methods are utilized for the estimation of kinetic parameters, as the E_a changes at various stages of conversion. The thermal behavior of TS, LDPE, 25%TS:75%LDPE, 50%TS:50%LDPE and 75% TS:25%LDPE during their pyrolytic and co-pyrolytic degradation can be further understood by estimating the E_a. However, this is done by incorporating the data to the TGA, by making the use of the model-free method. The E_a values are calculated from lines slope, as present in **Eqs. (4s-6s).** For the demonstration, only for the sample 50%TS:50%LDPE, linear fitting plots are presented in the supplementary data as Fig. 2**s(A-C).** The detailed kinetic parameters including the apparent E_a curve for pure TS and LDPE were illustrated in our previous work (Khan et al., 2026, 2025). The E_a values during TS pyrolysis varies between 23.06 kJ/mol to 205.10 kJ/mol for the Friedman, while for the KAS method the E_a values vary from 28.43 kJ/mol to 215.34 kJ/mol across different conversions. However, for the OFW the E_a values significantly vary from 21.83 kJ/mol to 40.06 kJ/mol. The E_a variation for TS occurs

Table 2
Weight loss during the co-pyrolysis of 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25%LDPE.

Samples	Stage	Weight loss (%)			
		2.5°C/min	5°C/min	7.5°C/min	10°C/min
TS	I	11.33	12.59	12.50	11.20
	II	24.60	23.81	23.72	23.41
	III	8.10	11.51	11.12	11.60
LDPE		100	100	100	100
25%TS:75% LDPE	I	10.02	8.04	8.0	9.03
	II	71.45	76.79	76	74.15
	III	3.55	2.85	2.9	3.38
50%TS:50% LDPE	I	8.48	5.37	5.99	6.83
	II	58.54	60.54	63.75	59.99
	III	6.62	5.23	4.59	5.97
75%TS:25% LDPE	I	8.12	7.88	7.63	7.20
	II	43.23	42.48	42.28	44.28
	III	8.01	8.51	8.1	8.34

because of its heterogenous composition and the degradation of TS sample. For the pyrolysis of LDPE significant variation of E_a values was observed. For the Friedman method the values are in the range of 244.72 kJ/mol to 301.16 kJ/mol across different conversions. Similarly for the KAS and the OFW the variation in E_a was observed between the range of 236.96 kJ/mol - 267.60 kJ/mol and 236.55 kJ/mol - 266.01 kJ/mol respectively at different conversions. This variation in E_a values arises due to chemical and the physical alteration of LDPE during the pyrolysis process (Zheng et al., 2018).

Similarly, Tables (1**s**-3**s**) in supplementary data shows the variation in E_a from 0.2 to 0.8 for all the blended samples. From Table 1**s** present in supplementary data for the 25%TS:75%LDPE, it can be seen that as conversion rises from 0.2 to 0.275, the E_a increase from the 259.12 kJ/mol - 262.19 kJ/mol for the Friedman, which is related to the volatiles release present in the 25%TS:75%LDPE. Moreover, the E_a values fluctuates from 0.3 to 0.8 conversion with E_a of 261.92 kJ/mol to 274.74 kJ/mol, indicating the shift of reaction mechanism because of the decomposition of various components and the chemical and physical properties alteration of LDPE in 25%TS:75%LDPE (Zheng et al., 2018).

Table 3
GC-MS analysis of pyro-oil obtained from 50%TS:50%LDPE.

Sr.	Compounds in pyro-oil	Molecular Formulae	Height (%)	Peak area (%)
Hydrocarbons and long chain aliphatic derivatives				
1	Long chain aliphatic compound	$C_{23}H_{44}O_3$	1.41	1.24
2	n-Octadecanoic acid, dimethyl (trimethylsilylmethyl)silyl ethyl	$C_{24}H_{52}O_2Si_2$	1.04	2.13
3	D-Mannitol, 1-decylsulfonyl-	$C_{16}H_{34}O_7S$	0.86	1.09
4	D-Glucitol, 4-O-decyl-	$C_{16}H_{34}O_6$	3.58	1.37
N- containing heterocycles and amines				
5	3-[N-Acetyl−4-acetylanilino] propionic acid	$C_{13}H_{15}NO_4$	0.73	1.13
6	Pyrazolo[1,5-*a*]pyrimidine−3-carbonitrile, 7 triflouromethyl−2,5-dimethyl	$C_{10}H_7F_3N_4$	1.35	1.45
7	Methyl[2-(1-methylpyrazol−4-yl) ethy]amine	$C_7H_{13}N_3$	1.41	2.59
8	4,6-Dimethyl−2-phenyl−4,5-dehydro−1,2-oxazine	$C_{12}H_{15}NO$	1.31	1.60
9	2-Methylaminomethyl−1,3-dioxolane	$C_5H_{11}NO_2$	1.44	1.44
Phthalates and Esters				
10	Bis(2-ethylhexyl) phthalate	$C_{24}H_{38}O_4$	74.35	74.00
11	Diglycolic acid, butyl 3,7-dimethyloctyl ester	$C_{18}H_{34}O_5$	1.49	1.42
Carboxylic and phenolic acids				
12	Benzoic acid, 3acetylamino−5-chloro−4-methyl, 2acetylaminoethyl ester	$C_{14}H_{17}ClN_2O_4$	0.89	0.98
13	3-Methylsalicylic acid, 2TMS derivative	$C_{14}H_{24}O_3SiO_2$	1.04	1.00
14	4-Imidazolidinecarboxylic acid, 4-hydroxy−2,5dioxo	$C_4H_4N_2O_5$	0.87	1.06
15	3-(4-Hydroxy−3-methoxyphenyl) propionic acid, 2TMS derivative	$C_{16}H_{28}O_4Si_2$	0.72	0.95
Cellulose and Carbohydrate derivative				
16	L-Gala-L-ido-octose	$C_8H_{16}O_8$	0.89	1.11
Cyclic ether and acetal				
17	1,3-Dioxane, 2-ethenyl−4,6-dimethyl-, (2 alpha, 4 alpha, alpha)	$C_8H_{14}O_2$	1.16	1.14
Silylated/TMS derivatives				
18	Phosphonates Acid, 3TMS derivative	$C_{11}H_{29}O_5PSi_3$	1.21	1.23
19	Trimethylsilyl-di(trimethylsilyl)-silane	$C_9H_{28}O_2Si_4$	3.07	1.80
20	3,3,5,5,7,7,9,9,11,11,13,13-Dodecamethyl−1,15-bis(1,3,3,5,5-pentamethyl−2,4,6-trioxa−1,3,5-trisilacyclohexyl0–4,6,8,10,12-tetraoxa−3,5,7,9,11-tetr	$C_{26}H_{74}O_{11}Si_{12}$	1.18	1.27
	Total		**100**	**100**

This also shows the various lumped reactions involve during the co-pyrolysis process (Wang et al., 2017). Additionally, volatiles from TS (often oxygenated compounds) react with the hydrocarbon chains from the LDPE which creates the synergistic effect that leads to the variation in E_a. The feasibility of these E_a for Friedman can be attributed to the high R^2 value of 1.00 at all conversions. Similarly for KAS method, variation in the E_a was observed across all conversions with the E_a value between 255.60 kJ/mol to 262.58 kJ/mol for KAS while for the OFW the E_a value increases from 0.2 to 0.8 conversion with E_a value of 254.22 kJ/mol to 261.35 kJ/mol. R^2 values of KAS and OFW lie between 0.97 and 1.00.

For 50%TS:50%LDPE from Table 2**s** present in supplementary data, the E_a value increases from 0.2 to 0.325 (65.63–168.59) kJ/mol. This increase in E_a occurs because of the complex degradation behavior during the initial stage. Furthermore, as conversion changes from 0.350 to 0.45, a decline in E_a is noted, with values decreasing from 163.04 kJ/mol to the 156.83 kJ/mol. However, the values of E_a again tend to increase from 0.475 to 0.8 conversion with the E_a value of 159.25 kJ/mol to 289.03 kJ/mol. The increase in the E_a was identified because of the increased fraction of TS in 50%TS:50%LDPE as it will increase the material complexity as compared to 25%TS:75%LDPE. The R^2 value for the Friedman method for 50%TS:50%LDPE lies between 0.99 and 1.00 that depicts the feasibility of E_a across all conversion ranges. Similarly, for 75%TS:25%LDPE shown in Table 3**s** which is present in supplementary data, it has been investigated that E_a value decreases from 0.2 to 0.3 conversion from 196.48 kJ/mol - 178.16 kJ/mol. The trend in E_a was opposite in comparison to both 25%TS:75%LDPE and 50%TS:50%LDPE. This is because degradation is mostly dominant by the TS as it contains moisture, volatiles, organic compounds and the oxygenated compounds which degrade normally at the low temperature and requires the low E_a. More to that as the conversion rises from 0.325 to 0.8 the apparent E_a will increase from 192.85 kJ/mol to 343.33 kJ/mol for Friedman method. However, both for the KAS and OFW, similar type of E_a trend was observed across these conversion ranges. The R^2 values for Friedman, KAS and OFW for the 75%TS:25%LDPE lie between 0.96 and 1.00, that indicates the E_a feasibility.

3.3.2. *Synergistic effect of co-pyrolysis parameters*

Fig. 5 presents the plot of temperature vs conversion difference obtained from experimental and theoretical values at the heating rate of 7.5 °C/min. The co-pyrolysis synergistic behavior of TS and LDPE demonstrates distinct interactions between these two materials. At the lower temperature around 0–200 °C, all exhibit the minimal conversion differences that indicate the limited interaction between the

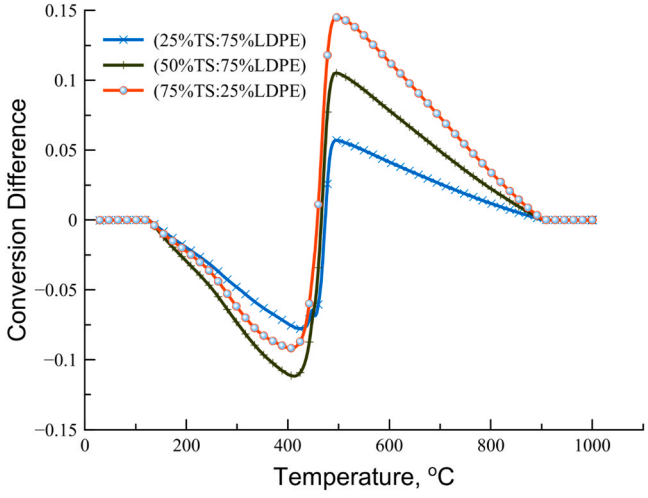


Fig. 5. Synergistic interactions during co-pyrolysis for 25%TS:75%LDPE, 50% TS:50%LDPE and 75%TS:25%LDPE.

components during the initial heating phase. As the temperature increases from 200 to roughly at 400 °C range, all the samples such as 25% TS:75%LDPE, 50%TS:50%LDPE, 75%TS:25%LDPE displayed the negative conversion differences, which shows the negative synergistic effects that suggests that combined materials have high theoretical values than expected based on their individual pyrolysis behaviors. The synergistic interaction will start increasing from 400 °C and the most significant positive synergistic behavior roughly occurs between 450 and 600 °C, where all the blends exhibit a sharp transition to positive conversion differences that indicate the beneficial interactions between TS and LDPE. Moreover, between 450 and 600 °C, the dominant peak occurs at 500 °C. The magnitude of this synergistic effect is strongly dependent on the blend composition. Prior literature shows that synergistic effect does not always vary consistently with the blending ratio of the samples (Wang et al., 2020b). The blend 75%TS:25%LDPE mixture achieved the maximum positive conversion difference of approximately 0.145 at 500 °C, followed by the 50%TS:50%LDPE blend that shows around 0.10, and the 25%TS:75%LDPE mixture showing around 0.055. This suggests that higher TS content causes more positive synergistic interactions during co-pyrolysis. This is highly attributed to the catalytic activity of the metals and inorganic minerals that are especially present in pure TS. The composition of pure TS is present in our previous published work (Khan et al., 2025). These inorganic minerals components can promote secondary cracking reactions and enhance the interaction between hydrogen-rich volatiles released from LDPE and oxygenated intermediates that somehow generates from TS. As a result, radical stabilization and conversion is improved. Furthermore, more experiments are needed to confirm this behavior. Beyond the peak temperatures, all blends show gradually declining conversion differences as the temperature increases towards 1000 °C and after that it eventually approaches to zero which indicates that the synergistic effects diminish at very high temperatures where organic components of both materials are largely decomposed.

4. Statistical analysis

4.1. Principle component analysis

Fig. 6 displays the PCA and temperature-dependent scoring results for the co-pyrolysis of TS and LDPE with varying blending ratios. Fig. 6 **(A)** shows the experimental biplot, which displays both the loadings and scores for the first two principal components such as PC1 and PC2 that are derived from 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25% LDPE. These components together cover 99.6% of the total variance in which PC1 alone accounts for the majority which indicates that the primary variations during co-pyrolysis are strongly influenced by LDPE content. The PC1 loadings show a strong positive correlation to LDPE-rich blends while the PC2 accounts for the small fraction of variance that somehow highlights the contributions primarily associated with the TS component which reflects its distinctive role in the process.

Fig. 6**(B)** shows the theoretical biplot with 99.2% variance with PC1 also strongly correlated with LDPE and PC2 more closely linked with TS. Fig. 6**(C)** compares theoretical and experimental PCA scores against the temperature. The blue solid line shows the theoretical scores which predict the expected behavior of the TS and LDPE blends. The red dashed line shows the experimental scores, which are actual measured data from the co-pyrolysis process. Fig. 6**(D)** shows the difference between the experimental and theoretical PCA scores against the temperature. Positive curve is observed roughly in the vicinity of 450–600 °C. This observation highlights regions where experimental results surpass theoretical predictions, suggesting enhanced reactivity, improved degradation efficiency, or the formation of novel chemical intermediates resulting from synergistic interactions between TS and LDPE. Conversely, negative troughs which are roughly in the temperature region of 200–400 °C indicate the temperature intervals where the co-pyrolysis process is suppressed or less effective than expected, possibly due to inhibitory interactions or competitive degradation pathways. To support our PCA findings, it can also be verified from Fig. 5 that positive synergistic interactions will appear around 450–600 °C range and the

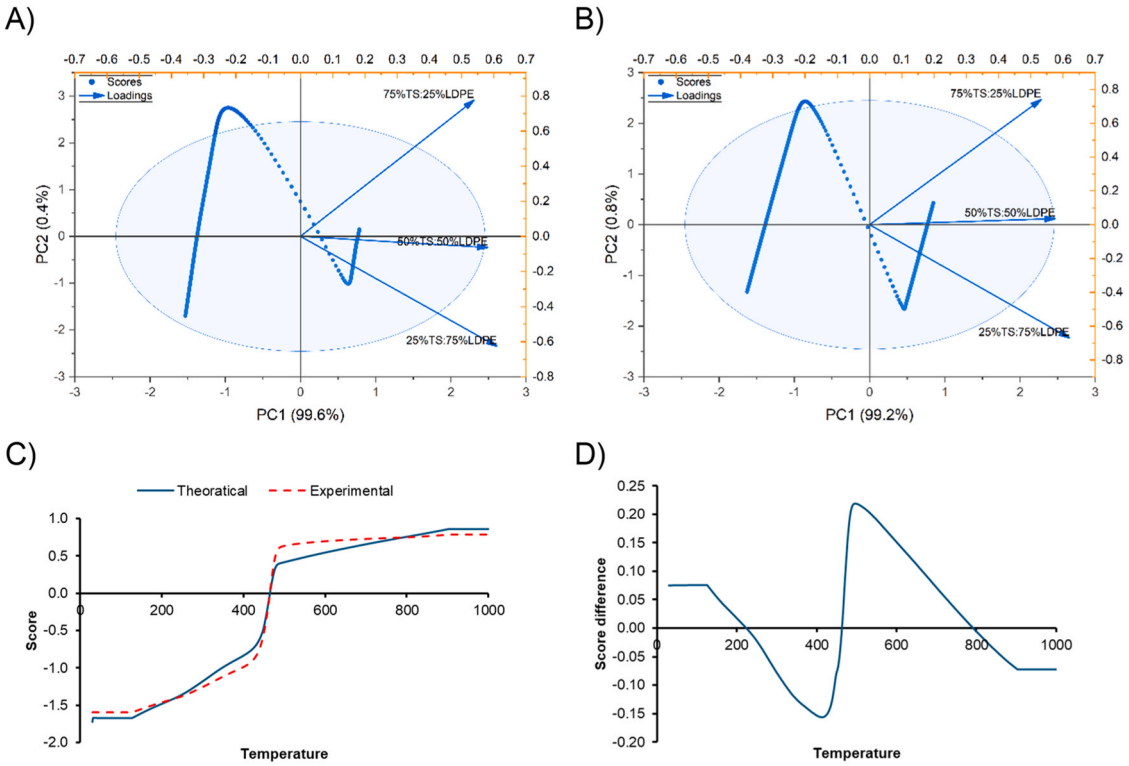


Fig. 6. Synergistic transformation of co-pyrolysis kinetic parameters **A)** experimental biplot, B) theoretical biplot, **C)** score plot, and **D)** score difference plot.

sharp maximum peak appears at the temperature of 500 °C. Between the temperature range of 450–600 °C, all the blends show a marked shift to positive conversion differences which indicate the favorable interactions between TS and LDPE at this temperature range.

4.2. Hierarchical cluster analysis

HCA was performed on the conversion data against the conversion rate obtained during the pyrolysis and co-pyrolysis of TS, LDPE, 25% TS:75%LDPE, 50%TS:50%LDPE, and 75%TS:25%LDPE. The results are visualized as a dendrogram (a tree-like diagram) shown in Fig. 3**s** in supplementary data. The result shows that 25%TS:75%LDPE, 50% TS:50%LDPE, 75%TS:25%LDPE positioned close together on the dendrogram, which suggests they have similar co-pyrolysis characteristics.

5. Product analysis from co-pyrolysis

The sample 50%TS:50%LDPE was chosen for co-pyrolysis process in a reactor (fixed bed) because of the profound results of apparent E_a among the blended samples, as the average E_a value from Friedman method is relatively lower compared to other blended samples such as 25%TS:75%LDPE and 75%TS:25%LDPE. The co-pyrolytic breakdown of 50%TS:50%LDPE was carried out at 500 °C and the pressure of nitrogen was adjusted as 0.2 MPa. The fixed bed reactor was totally sealed from all sides, and the soap bubble test was carried out to ensure there were no gas leaks. A total of 8 g of the sample was placed in center of the reactor. The resulting pyro-oil obtained as a dark brown substance from the co-pyrolysis process. The yields from this co-pyrolysis were approximately 22% for pyro-oil, 34% for biochar, and 44% for non-condensable gases. In the previous work yield of biochar obtained from sewage sludge and HDPE co-pyrolysis at 500 °C was found 23.5% (Ma et al., 2024). Furthermore, copyrolysis of macroalgae (Ulva. Lactuca) and polyethylene terephthalate (PET) was also investigated where the pyro-oil yield was highest at 37.91% at 500 °C for (40%U.lactuca:60%PET) (Farobie et al., 2024).

Fig. 7**(A)** identifies the GC-TCD analysis results from co-pyrolysis which shows the volumetric percentages of non-condensable gases. The co-pyrolysis process produces the volatile compounds both from the initial breakdown of the material and from the further decomposition of vaporized substances. The results indicate the formation of light gases such as hydrogen (H_2) and methane (CH_4), which remain gaseous at high temperatures. Initially, hydrogen concentration is about 0.45 vol%, while methane is at 1.19 vol%, attributed to heavier hydrocarbons and tars that resist the hydrogen release. After 20 min, the levels of H_2 and the CH_4 rise to approximately 1.8 and 3.2 vol%, respectively, due to intensified secondary reactions enhancing CH_4 production (Zhang et al., 2011). As time continues to increase, both the H_2 and CH_4 concentrations decrease to roughly 1.1 vol% and 1.6 vol%, respectively. With further processing time, H_2 declines from 1.1 vol% to 1 vol%, and CH_4 from 1.6 vol% to 1.3 vol%. However, toward the end, hydrogen slightly recovers from 1 vol% to 1.2 vol%. Additionally, it is identified by the previous literature that different oxides of carbon (CO and CO_2) and some other light hydrocarbons are also produced during the co-pyrolytic degradation because of carboxyl groups (Liu et al., 2018). The non-condensable gases concentration were calculated based on a mix of complex reactions which includes the heterogeneous reactions such as the water-gas reaction, hydrogasification, and the Boudouard reaction, along with the homogeneous reactions like reforming and methanation reactions (Leung et al., 2002).

The GCV of 50%TS:50%LDPE derived pyro-oil by co-pyrolysis was obtained as 41.579 MJ/kg. In our prior work, the GCV of pure TS pyro-oil and LDPE pyro-oil was 20.067 MJ/kg and the 40.694 MJ/kg (Khan et al., 2026, 2025). The slightly higher GCV of 50%TS:50%LDPE pyro-oil as compared to LDPE pyro-oil was possibly due to synergistic interaction and the cracking which can shift the pyro-oil composition

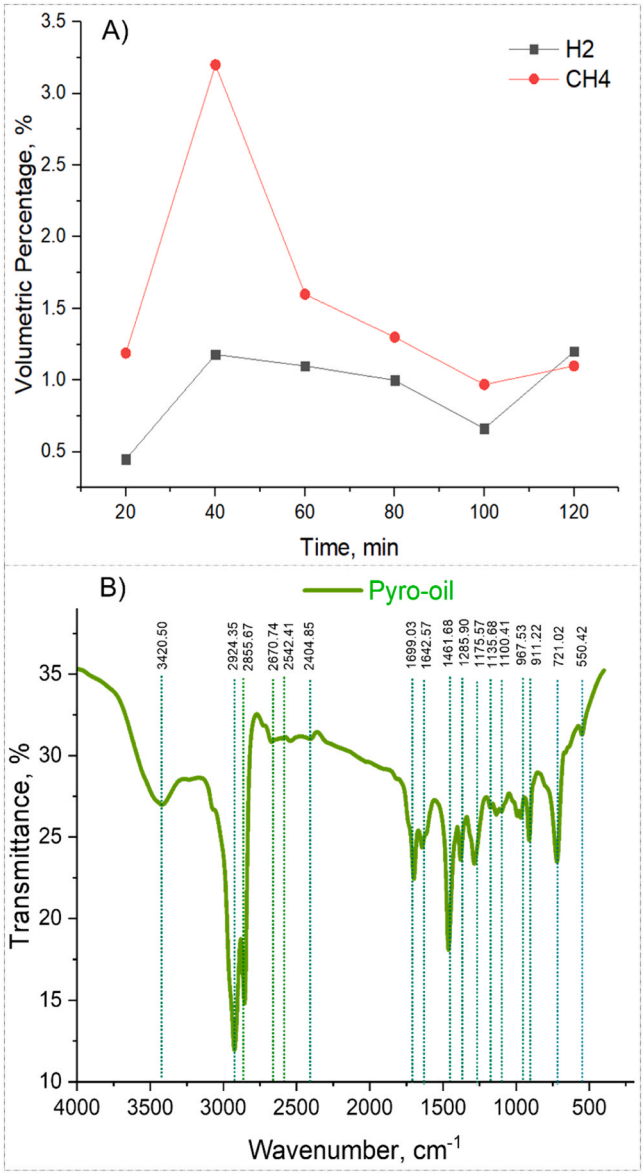


Fig. 7. (A) non-condensable gases **(B)** pyro-oil obtained from co-pyrolysis of 50%TS:50%LDPE.

toward long chain aliphatic hydrocarbons. This indicates that pyro-oil obtained from 50%TS:50%LDPE is energy dense and of high quality. In the previous literature co-pyrolysis of lignocellulosic biomass with plastic was studied where the co-pyrolysis of blends improves the heating value of pyro-oil (Vo et al., 2022). In another the prior literature, co-pyrolysis of food waste with plastic waste was also investigated where the co-pyrolysis improved the heating value of pyro-oil to 42.5 MJ/kg for 50:50 blend for food waste and plastic waste because of the significant synergistic interactions between the two feedstocks (Amrullah et al., 2022). FTIR analysis of the pyro-oil produced from the co-pyrolysis of reveals significant information about the chemical composition and functional groups present, which is shown in Fig. 7**(B).** The FTIR spectra shows prominent peaks corresponding to various functional groups in pyro-oil. A broad peak roughly in the range of 3200–3420.50 cm^{-1} indicates the presence of hydroxyl (O-H) groups, suggesting the formation of alcohols and phenols (Liu et al., 2018). Peak at 1699.03 cm^{-1} is associated with carbonyl (C═O) groups, indicating the presence of ketones, aldehydes, or carboxylic acids. Additionally, peaks in the region of 2855.67–2924.35 cm^{-1} correspond to C-H stretching vibrations, that somehow highlights the presence of alkanes

and alkenes, which are indicative of hydrocarbon structures in the pyro-oil. The FTIR spectrum may also show peaks around 1461.68–1642.57 cm^{-1}, which are characteristic of aromatic compounds. This suggests that the co-pyrolysis process effectively converts the 50%TS:50%LDPE mixture into complex aromatic hydrocarbons. The diverse hydrocarbon structure is the representative of the higher GCV of this pyro-oil produced from co-pyrolysis.

Gas Chromatography-Mass Spectrometry (GCMS) results for pyro-oil obtained by co-pyrolysis of 50%TS:50%LDPE at 500 °C reveals detailed information about the chemical compounds. Pyro-oil obtained by thermochemical process such as pyrolysis is a complex mixture with diverse chemical functionalities, and its composition is influenced by the type of biomass that is used and the conditions under which it is processed (Sahoo et al., 2021). Typically, pyro-oil, obtained from non-lignocellulosic materials like sludge, contains the nitrogenous and oxygenated compounds (Ma et al., 2024; Pokorna et al., 2009). Specifically, pyro-oil from 50%TS:50%LDPE features a variety of organic compounds which are classified into different groups. The major components identified in the pyro-oil include the various aromatic along with the aliphatic hydrocarbons and their derivatives, which enhances the pyro-oil energy content and flammability. Additionally, ketones and carboxylic acids presence indicates that the pyro-oil derived from co-pyrolysis also comprises oxygen containing compounds, which impact reactivity and the potential applications, such as fuel and the chemical feedstock. Similar class of the compounds have been presented in literature concerning pyrolytic oil from the raw TS (Scheibe et al., 2022). Furthermore, esters have also been found in the pyro-oil. Studies indicate that at the pyrolysis temperatures between 400 and 600 °C, the resulting oil may exhibit both aromatic and aliphatic structures along with oxygenated species. Generally, hydrocarbons are valuable due to their industrial significance as additives and chemicals that improve fuel octane ratings (Garcia-Perez and Metcalf, 2008). Moreover, the pyrolysis of sludge containing long-chain alkanes results in the formation of arenes through processes such as alkane dehydrogenation to the alkenes and dienes, followed by aromatization and cyclization (Huang et al., 2014). If condensation reactions persist, highly condensed polycyclic compounds may form. Table 3 presents the results obtained from the GC-MS analysis of pyro-oil obtained from the co-pyrolysis of 50%TS:50%LDPE. The predominant compound identified as Bis(2-ethylhexyl) phthalate, that is a type of the organic compound known as a phthalate ester, which accounted for 74% of the peak area with 74.35% height. The presence of phthalate compound is likely because of the presence of additives or plasticizers in the LDPE feedstock. This is originally incorporated in the LDPE material during processing. Furthermore, presence of phthalate is also attributed to the thermal degradation of oxygen rich constituents in TS. In the previous literature, phthalate is obtained in pyro-oil and the phthalate-containing pyro-oil fractions can still be considered for fuel applications when blended or utilized as part of complex fuel mixtures (Hossain et al., 2019). Another significant compound detected was 2-Methyl-aminomethyl−1,3-dioxolane, which are a class of organic compounds. Additionally, Benzoic acid, 3-acetylamino−5-chloro−4-methyl, −2acetylyaminoethyl ester, were observed with a peak area of 0.98%. The benzoic acid structure in this compound depicts the aromaticity. It can also be observed from Table 3 that various N-heterocyclic compounds are also detected such as Pyrazolo[1,5-*a*] pyrimidine−3-carbonitrile, 7 triflouromethyl−2,5-dimethyl and 4,6-Dimethyl−2-phenyl−4, 5-dehydro−1,2-oxazine. Compound such as Methyl[2-(1-methylpyrazol−4-yl) ethyl] amine is also a nitrogenous compound with potential applications in chemical synthesis. Moreover, the compound 3-[N-Acetyl−4-acetylanilino] propionic acid contains the aniline derivative, which is also N-heterocycle compound. Furthermore, n-Octadecanoic acid, dimethyl(trimethylsilylmethyl)silyl ethyl with a peak area of 2.13% contains the hydrocarbon and long chain aliphatic derivative. The dominant compound of ester in the co-pyrolysis pyro-oil was found Diglycolic acid, butyl 3,7-dimethyloctyl ester with a peak area of 1.42. Fig. 8 presents the schematic representation of the chemical pathways, along with the products derived from the co-pyrolysis of TS with LDPE, highlighting the formation of various organic compounds through degradation, cross-reactions, and cracking processes.

GCV of the received biochar from 50%TS:50%LDPE co-pyrolysis was found to be ~5.981 MJ/kg. The SEM analysis of the biochar produced from the co-pyrolysis of a 50%TS:50%LDPE at 500 °C reveals the complex and varied surface morphology which is shown in Fig. 9**(A-D).** The SEM images shows a rough texture with a mix of charred particles and the pore structure becomes compact and well aligned in regular pattern (Zhang et al., 2020a). This is attributed to the decomposition and volatilization of the organic matter (Zhang et al., 2020b).

The FTIR analysis of the 50% TS:50% LDPE blend derived from biochar is shown in Fig. 9**(E).** The spectrum generally shows a reduction in oxygen-containing functional groups due to the high co-pyrolysis temperature of 500 °C, which promotes deoxygenation and dehydration reactions. The peak at 3426.54 cm^{-1} represents the O-H stretching vibrations of the hydroxyl groups. This peak appears at 3429.95 cm^{-1} for raw 50%TS:50%LDPE with sharp peak. The decrease in peak results from the breakdown of significant quantities of free and associated hydroxyl groups, as well as the structural hydroxyl groups (−COOH and −COH), present in the biochar sample (Sabri et al., 2021).

Peaks near 2853.58 cm^{-1} correspond to O-H stretching vibration of the carboxyl group. The shoulder at 1631.80 cm^{-1} shows the presence of

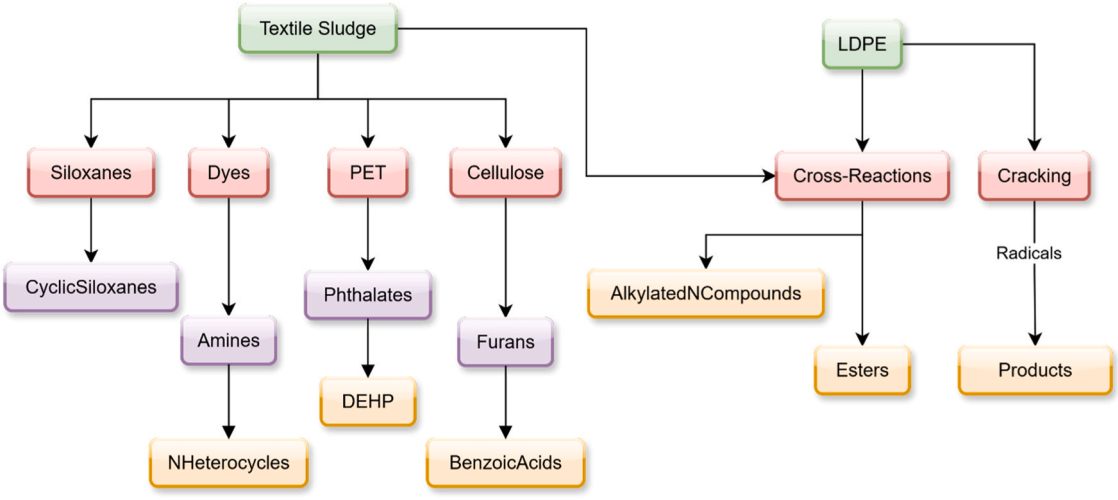


Fig. 8. Plausible co-pyrolysis product pathway from 50%TS:50%LDPE during degradation and reaction processes.

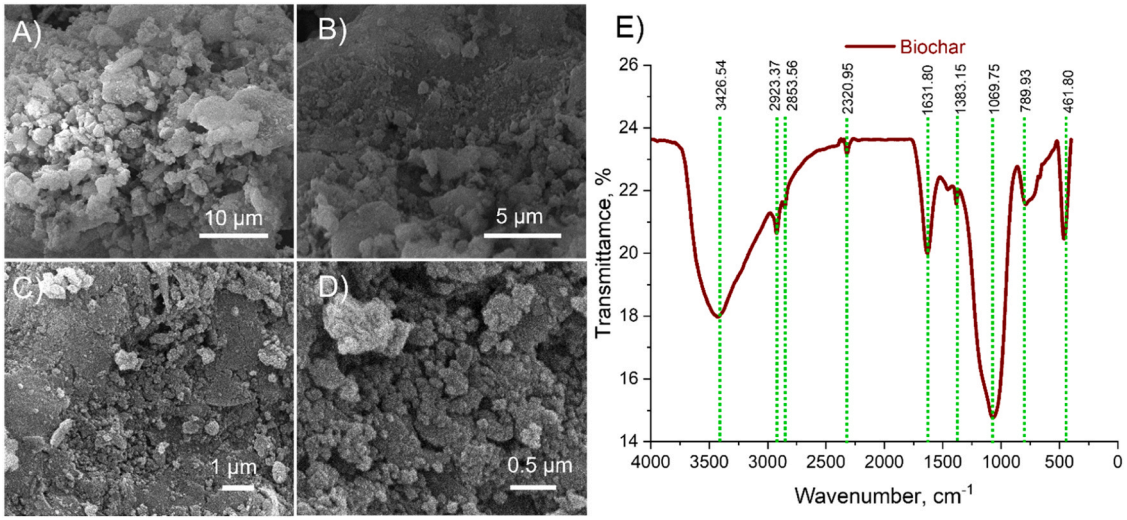


Fig. 9. 50%TS: 50%LDPE biochar **(A-D)** SEM images & **(E)** FTIR Spectrum.

amide bond stretching (CO=NH). It has also been observed that the characteristic peak shifted from 1654.03 cm^{-1} for raw 50%TS:50% LDPE to 1631.80 cm^{-1}, which shows that the amide functional group might have complexed with the heavy metal present in the sludge sample. The sharp peak around 1069 cm^{-1} C-O shows the stretching vibration and the O-H deformation, while the peak around 1300 cm^{-1} indicates the S=O stretching vibration (Sabri et al., 2021) suggesting the presence of ketones or carboxylic acids.

6. Machine learning assessment

Different models like ANN, C&RT, SVM were applied to the blended samples such as 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25% LDPE to predict the E_a. These predictive models compare the computed E_a with those of the actual E_a that facilitate the predictive model's formation. As illustrated in Fig. 10**(A, C, E)**, Fig. 11**(A, C, E)** and Fig. 12**(A, C, E)** all the models which includes ANN, C&RT and the SVM demonstrated a strong correlation with E_a values derived by Friedman method for the 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25%LDPE. Additionally, Fig. 10**(B, D, F),** Fig. 11**(B, D, F)** and Fig. 12**(B, D, F)** shows that for 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25% LDPE all predictive models like ANN, C&RT, and SVM align closely with values of E_a that is obtained by Friedman's for all heating rates, especially at lower values of E_a. Additionally, E_a values obtained through the Friedman method for 25%TS:75%LDPE ranged from 258.57 to 274.74 kJ/mol, while for the 50%TS:50%LDPE the E_a values are in between 65.63 and 289.03 kJ/mol within a conversion range of 0.2–0.8. Additionally, for the sample 75%TS:25%LDPE the Friedman E_a values lie between 178.16 and 343.33 kJ/mol. In comparison, the different machine learning models exhibited varying ranges of E_a. For 25% TS:75%LDPE the produced the E_a values lie between 258.8 and 273.64 kJ/mol, 258.12–274.30 kJ/mol and 259.39–271.43 kJ/mol, respectively for ANN, C&RT and SVM. While for the 50%TS:50%LDPE the E_a values of these models are in the range of 65.14–287.66 kJ/mol, 71.21–273.39 kJ/mol and 55.55–265.39 kJ/mol, respectively. More to that for 75%TS:25%LDPE the produced E_a values for ANN, C&RT and SVM lie in the range of 177.80–337.18 kJ/mol, 179.06–338.06 kJ/mol and 175.06–329.88 kJ/mol, respectively. As shown in Table 4**,** it is confirmed that the results from all the developed models show general consistency with E_a values obtained by Friedman method. Table 5 presents the performance matrix for all the developed models during the co-pyrolysis process.

The performance of the various models during the co-pyrolysis process was evaluated using ANN, C&RT and SVM focusing on several performance metrics. From the Table 5, it can be seen that for 25% TS:75%LDPE the C&RT model achieved an impressive R^2 value of 0.988, making it the top performer among all other predictive models, having the R^2 values of 0.942 and 0.837 for ANN and SVM, respectively. However, SVM has the highest RMSE of 1.442 while C&RT has the lowest RMSE with a value of 0.381. Similarly, for the sample 50% TS:50%LDPE, ANN outperformed C&RT and SVM with a higher R^2 of 0.999 and having the lowest RMSE of 0.873. However, the R^2 values of other predicted models, like C&RT and SVM are found to be 0.995 and 0.934 and RMSE of 2.814 and 11.112, respectively. Notably, the ANN not only had the highest R^2 but also the lowest MSE of 0.762, which shows its superior predictive capability for E_a during co-pyrolytic degradation. Furthermore, the absolute mean error (AME) was assessed for all models, with ANN showing AME of 0.619, while the C&RT AME was obtained as 1.626. However, SVM exhibited AMEs of 8.755, respectively. The combination of high R^2 values and low error demonstrates that these models can precisely predict the E_a with significant precision. Furthermore, for the sample 75%TS:25%LDPE the ANN has the most important predictive capabilities having a high R^2 value and low RMSE. The R^2 value was found to be 0.997 and the RMSE of the ANN was observed as 2.158. The R^2 values of C&RT and SVM were observed as 0.995and 0.973 respectively.

7. Conclusion

Co-pyrolysis of TS with LDPE has emerged as a notable solution for energy production that leads to several important conclusions:

- The study successfully utilized model-free kinetics which are vital for clarifying the co-pyrolysis kinetic parameters of 25%TS:75%LDPE, 50%TS:50%LDPE, and 75%TS:25%LDPE.
- Co-pyrolyzing TS and LDPE predominantly influence the synergistic interaction and the significant positive synergistic interactions were observed between the temperature of 450–600 °C with the maximum peaks appear at 500 °C which indicates the enhanced reactivity and favorable interactions in this range.
- 50%TS:50%LDPE was selected for product analysis based on the lower value of average E_a of the Friedman. The co-pyrolysis products yielded the pyro-oil constitutes to about 22%, non-condensable gases make up 44%, and biochar represents 34%. These findings provide critical data for evaluating the efficiency and economic feasibility of the co-pyrolysis of 50%TS:50%LDPE. The pyro-oil composition which contains diverse long chain aliphatic hydrocarbons along with

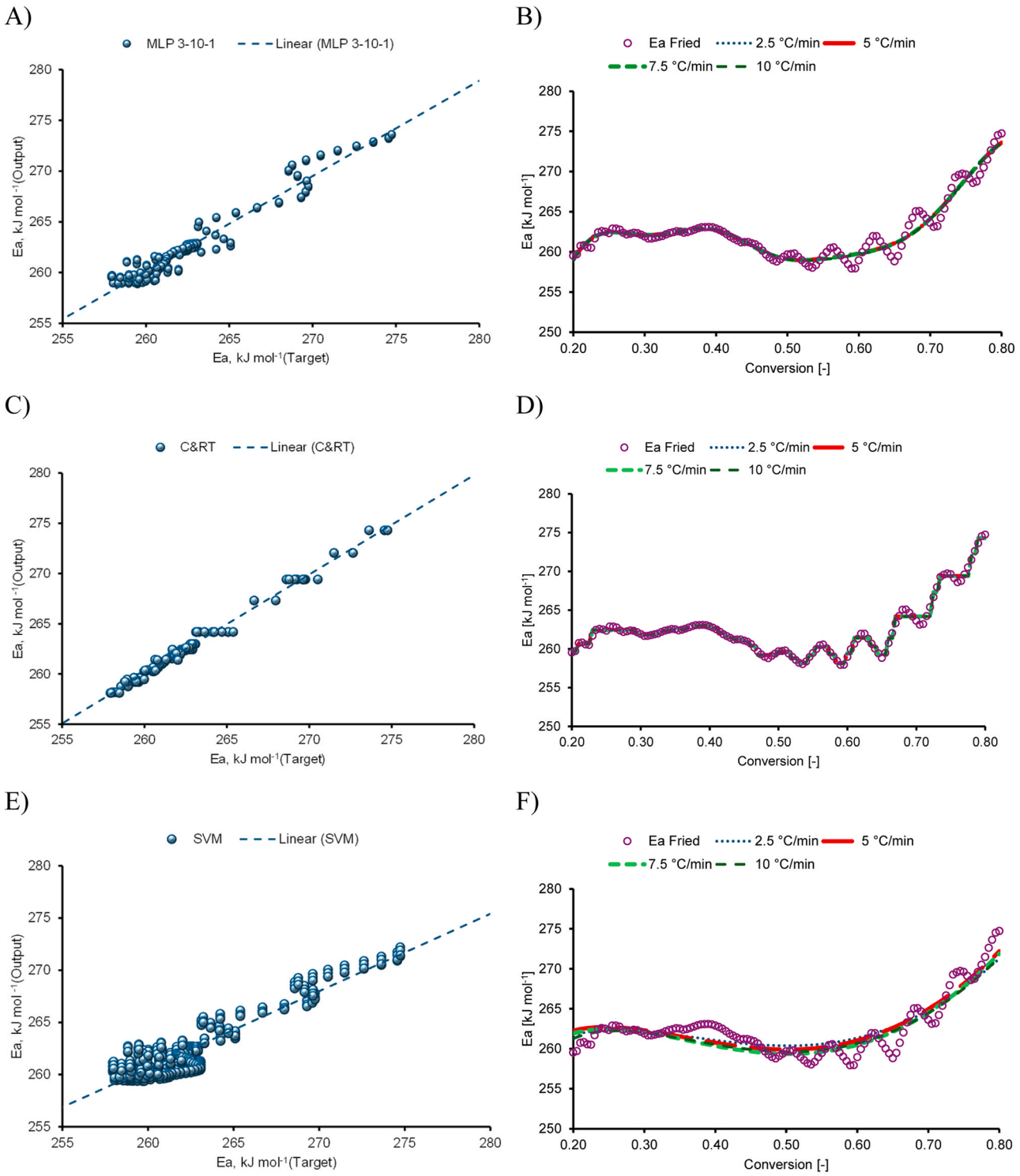


Fig. 10. Correlation along with prediction plots for the best fit models ANN **(A&B),** C&RT **(C&D)** and SVM **(E&F)** models for the E_a of 25%TS:75%LDPE co-pyrolysis.

other compounds suggests its potential use in energy generation and in chemical feedstocks.

- The integration of machine learning predictive models, including ANN, C&RT, and SVM significantly enhances predictive accuracy regarding the thermal degradation behavior of co-pyrolysis process.
- C&RT outperformed other models like ANN and SVM in terms of predictive accuracy for 25%TS:75%LDPE, while ANN has the superior predictive capabilities over C&RT and SVM for 50%TS:50% LDPE and for 75%TS:25%LDPE.

Future research on the co-pyrolysis of TS and LDPE should prioritize the development of advanced catalysts to improve the yield of valuable co-pyrolysis products. Furthermore, conducting the comprehensive life cycle assessment would be crucial for evaluating the viability of scaling the co-pyrolysis process for industrial applications.

CRediT authorship contribution statement

Imtiaz Ali: Writing – review & editing, Investigation, Data curation, Conceptualization. **Waheed Miran:** Writing – review & editing,

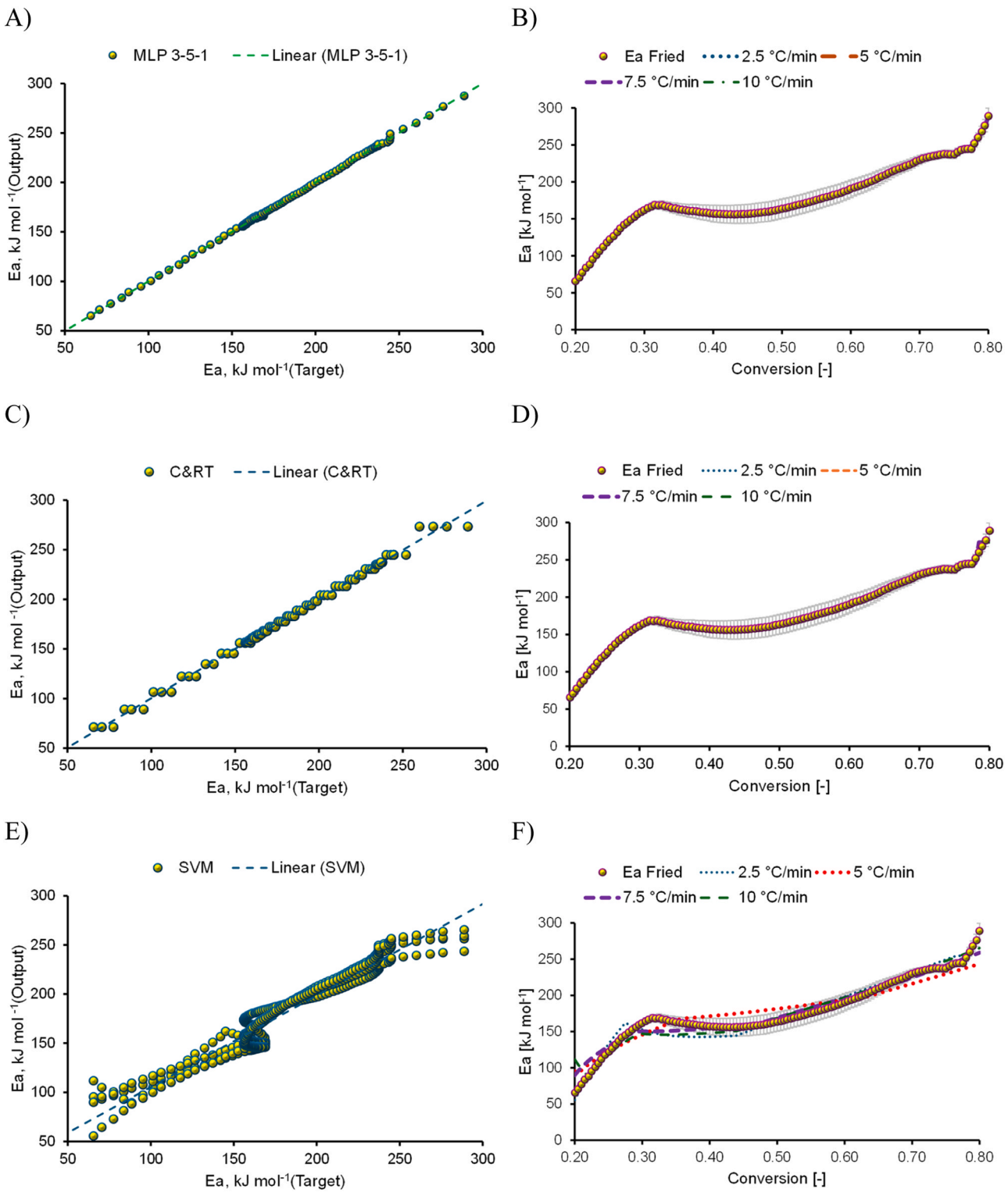


Fig. 11. Correlation along with prediction plots for the best fit models ANN**(A&B),** C&RT **(C&D)** and SVM **(E&F)** models for the E_a of 50%TS:50%LDPE co-pyrolysis.

Supervision, Resources, Conceptualization. **Erum Pervaiz:** Writing – review & editing, Formal analysis, Conceptualization. **Naqvi Salman:** Writing – review & editing, Formal analysis, Conceptualization. **Dagmar Juchelková:** Writing – review & editing, Formal analysis, Conceptualization. **Khoja Asif:** Writing – review & editing, Resources, Methodology, Conceptualization. **Arslan Khan:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

LLMs were used to enhance language and readability. The authors reviewed and edited the content and take full responsibility for the publication.

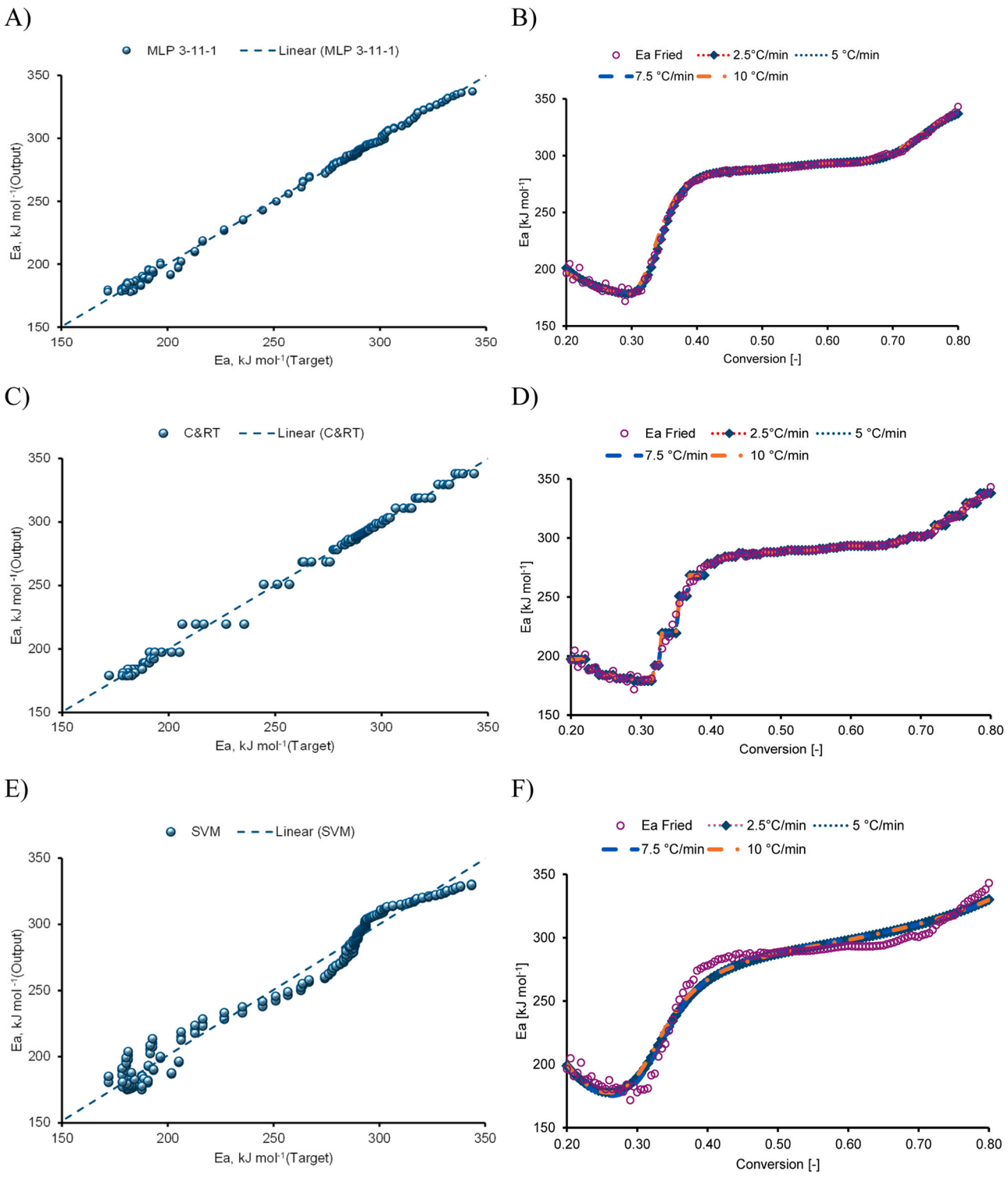


Fig. 12. Correlation along with prediction plots for the best fit models ANN **(A&B),** C&RT **(C&D)** and SVM**(E&F)** models for the E_a of 75%TS:25%LDPE co-pyrolysis.

Table 4
E_a range for Friedman and Predicted E_a of 25%TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25%LDPE from 0.2 to 0.8 conversion.

Sample	Friedman, E_a (kJ/mol)	Predicted E_a ANN (kJ/mol)	Predicted E_a C&RT (kJ mol)	Predicted E_a SVM (kJ/mol)
25%TS:75%LDPE	258.57–274.74	258.8–273.63	258.12–274.30	259.39–271.43
50%TS:50%LDPE	65.63–289.03	65.14–287.66	71.21–273.39	55.55–265.39
75%TS:25%LDPE	178.16–343.33	177.80–337.18	179.06–338.06	175.06–329.88

Table 5
Machine learning performance metrics for predicting co-pyrolysis E_a of 25% TS:75%LDPE, 50%TS:50%LDPE and 75%TS:25%LDPE.

Sample	Algorithm	AME	MSE	RMSE	R^2
25%TS:75%LDPE	ANN	0.628	0.741	0.861	0.942
	C&RT	0.276	0.145	0.381	0.988
	SVM	1.140	2.079	1.442	0.837
50%TS:50%LDPE	ANN	0.619	0.762	0.873	0.999
	C&RT	1.626	7.919	2.814	0.995
	SVM	8.755	123.490	11.112	0.934
75%TS:25%LDPE	ANN	1.452	4.659	2.158	0.997
	C&RT	1.735	9.576	3.094	0.995
	SVM	6.580	60.603	7.784	0.973

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.cherd.2026.06.007.

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